Scaling Up Electrochemical CO_2 Reduction: Enhancing Performance of Metalloporphyrin Complexes in Zero-Gap Electrolyzers

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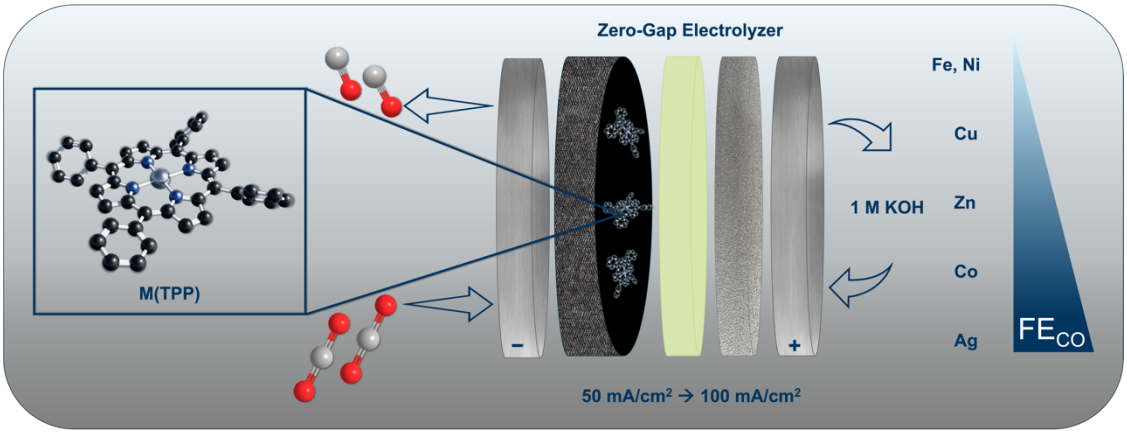
Molecular Catalysis

In recent years, the immobilization of molecular complexes onto carbon supports for electrochemical CO_2 reduction has gained significant attention. Metalloporphyrins, in particular, have often been studied when immobilized on carbon nanotubes for electrolysis in laboratory-scale systems, such as H-type cells. However, there is considerable interest in scaling these molecular complexes beyond the lab and integrating them into industrially applicable electrolyzer types, such as zero-gap cells. Compared to pure metal (nano)particles, transition metal complexes often allow for lower overall metal loadings, leading to a higher mass activity while maintaining a comparable performance. Therefore, implementing metalloporphyrins in such application-oriented electrolyzers is of great interest. In this study, six different tetraphenylporphyrin (TPP, M = Fe, Co, Ni, Cu, Zn, Ag) complexes were incorporated into gas diffusion electrodes and applied in a zero-gap electrolyzer.[1] In addition to examining the influence of the metal center, the impact of the carbon support was also investigated. Most complexes performed best in the presence of carbon black, but for instance, Co(TPP) showed a better performance with CNTs, demonstrating that the choice of support material can significantly enhance catalytic performance. The faradaic efficiencies (FE) for the reduction product CO varied considerably depending on the metal center, with CO generation FEs ranging from less than 1% (Fe(TPP), Ni(TPP)), over 37% (Zn(TPP)) and 75% (Co(TPP) to nearly 100% (Ag(TPP)) at an applied current density of 50 mA/cm². Wherein it is especially interesting, that the homogenously broadly studied and highly active Fe(TPP) loses its activity under the applied conditions, which can be attributed to decomposition under the high current densities. Additionally, methane generation with an FE of 6% was observed when the

Cu(TPP) complex was implemented into a GDE. Even when the current density was doubled to 100 mA/cm², the FE for CO of Ag(TPP) only decreased to 92% at a cell voltage of 3 V. This work highlights on the one hand the potential of molecular systems for scaling up to industrial applications and on the other that a testing in application orientated conditions is necessary to draw conclusions on suitable catalysts for scale up.



[1] W. Wiesner, J. Y. Maldonado Arias, J. Jökel, R. Cao and U.-P. Apfel, *Chem. Commun.*, **2024**, in revision.